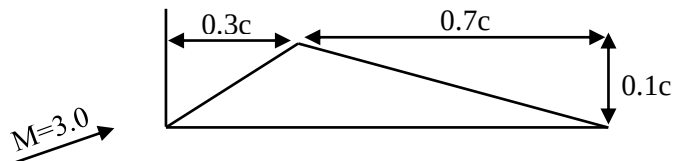


1-Consider the two-dimensional airfoil shown in the figure with a freestream Mach number of 3.0 and an angle of attack of 2° . The ratio of thickness to chord is 0.1 and the maximum thickness occurs at a distance of 30% of the chord from the leading edge. Compute the pressure ratio for each surface, the lift coefficient, and the drag coefficient using the shock/expansion method and linearized theory.



2- Consider the following diamond-shaped airfoil at a Mach number of 2.0 in a wind tunnel. The wind tunnel conditions are as follows: $p=125$ psi, $T=650^\circ\text{R}$. The maximum thickness is $0.4c$, and the angle of attack is $\alpha=10^\circ$. Find the pressure in all regions, the lift, and the drag coefficients using the shock/expansion method and linearized theory. $(\delta_2 = 8 \text{ and } \delta_3 = 2)$

